

Daily Problem: 26–Feb–2026

Problem Statement

Show that any deterministic k -tape Turing machine M that runs in time $T(n)$ on inputs of length n can be simulated by a single-tape Turing machine S in time $O(T(n)^2)$.

Proof

Let M be a deterministic k -tape TM. On any input x of length n , suppose M halts within $T(n)$ steps.

We construct a single-tape TM S that simulates M step-for-step, with total running time $O(T(n)^2)$.

Step 1: Bounding the used tape region

In $T(n)$ steps, each tape head of M can move at most $T(n)$ cells away from its start position. Therefore, for each of the k tapes, all nonblank cells ever visited/modified by M lie within a window of length $O(T(n))$. So the total relevant information across all tapes fits within $O(kT(n)) = O(T(n))$ symbols (since k is a fixed constant).

Step 2: Encoding all k tapes onto one tape

Machine S keeps on its single tape an encoding of the k tapes of M , separated by delimiters. For example, it maintains a string of the form

$$\# w_1 \# w_2 \# \cdots \# w_k \#,$$

where w_i is the current content (within the relevant window) of tape i of M .

To record head positions, we mark the symbol under each head. One convenient way is to use a *dotted* version of the tape alphabet. That is, if Γ is the tape alphabet of M , then S uses an alphabet $\Gamma' = \Gamma \cup \dot{\Gamma} \cup \{\#\}$, where $\dot{a}$ indicates that the head is currently scanning symbol a .

Thus, for each block w_i , exactly one symbol is dotted, indicating the head position on tape i .

Step 3: Simulating one step of M

Suppose at some point S holds a correct encoding of the full k -tape configuration of M . We show how S simulates one transition of M .

A single step of M depends on:

- the current state q of M , and
- the k symbols currently under the k heads.

So S must:

1. **Read the k scanned symbols.** S makes one left-to-right pass over its tape to find the dotted symbols in each block w_i , thereby learning the k scanned symbols. It stores these k symbols (and the state q) in its finite control.
2. **Apply M 's transition rule.** Since M is deterministic, the transition function determines:
 - the new state q' ,
 - the k symbols to write (replacing the scanned ones),
 - and the direction each head moves (Left/Right/Stay).

This computation is done in S 's finite control.

3. **Update the encoding.** S makes a second pass over the tape. When it reaches each dotted symbol, it:
 - replaces it with the new written symbol (dotted or undotted as appropriate),
 - moves the dot one cell left or right according to the head movement direction.

If moving the head would go beyond the currently represented window, S extends that block by inserting blanks at the boundary.

Step 4: Time for one simulated step

At any time during the simulation, the total length of the encoded tape on S is

$$O(kT(n)) = O(T(n)).$$

Each simulated step of M requires S to make a constant number of full passes over this encoding (two passes suffice in the description above). Each pass takes $O(T(n))$ time. Therefore, each simulated step costs $O(T(n))$ time on S .

Step 5: Total running time

Machine M takes at most $T(n)$ steps. For each of these steps, S spends $O(T(n))$ time.

Hence the total time is

$$T(n) \cdot O(T(n)) = O(T(n)^2).$$

Conclusion

A deterministic k -tape TM running in time $T(n)$ can be simulated by a single-tape TM in time $O(T(n)^2)$, by encoding all tapes on one tape and using $O(T(n))$ -time scans to simulate each step.